

PROGRAM TO FIND COMMON ITEMS IN TWO LIST

```
list1 = input("Enter elements of first list separated by spaces: ").split()
list2 = input("Enter elements of second list separated by spaces: ").split()
common = False
for item in list1:
    if item in list2:
        common = True
        break
print("Have common member:", common)
```

PROGRAM TO COUNT NO OF LINES IN A FILE

```
filename = input("Enter the file name: ")
with open(filename, "r") as f:
    line_count = sum(1 for line in f)
print("Number of lines:", line_count)
```

PRINT LINES CONTAINING CONSECUTIVE 1'S

```
filename = input("Enter the file name: ")
with open(filename, "r") as f:
    lines = f.readlines()
for line in lines:
    if "11" in line:
        print(line.strip())
```

PRORAM TO FIND WHICH DATE IS EARLIER

```
date1 = input("Enter the first date (day month year): ").split()
date2 = input("Enter the second date (day month year): ").split()
# Convert each value separately
day1 = int(date1[0])
month1 = int(date1[1])
year1 = int(date1[2])
day2 = int(date2[0])
month2 = int(date2[1])
year2 = int(date2[2])
```

```
# Store dates as (year, month, day)
d1 = (year1, month1, day1)
d2 = (year2, month2, day2)
# Compare the dates
if d1 < d2:
    print("The first date is earlier.")
elif d2 < d1:
    print("The second date is earlier.")
else:
    print("Both dates are the same.")
```

MATRIX OPERATION USING NUMPY

```
import numpy
matrix1 = numpy.matrix([[1, 2, 3], [1, 1, 1], [1, 1, 1]])
matrix2 = numpy.matrix([[1, 0, 0], [0, 1, 0], [0, 0, 1]])
print("Matrix 1=\n", matrix1)
print("Matrix 2=\n", matrix2)
matrix3 = numpy.add(matrix1, matrix2)
matrix4 = numpy.subtract(matrix1, matrix2)
matrix5 = numpy.matmul(matrix1, matrix2)
matrix6 = numpy.transpose(matrix1)
print("Matrix 1 + Matrix 2=\n", matrix3)
print("Matrix 1 - Matrix 2=\n", matrix4)
print("Matrix 1 * Matrix 2=\n", matrix5)
print("2 * Matrix 1=\n", 2 * matrix1)
print("Transpose of Matrix 1=\n", matrix6)
```

HISTOGRAM OF STUDENTS MARK

```
import matplotlib.pyplot as plt
# Given marks
marks = [22, 87, 5, 43, 56, 73, 55, 54, 11, 20, 51, 5, 79, 31, 27]
# Define bins for the intervals (0–10, 10–20, ..., 90–100)
bins = [0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
```

```
# Create histogram
plt.hist(marks, bins=bins, edgecolor="black")
# Labels and title
plt.xlabel("Marks Range")
plt.ylabel("Number of Students")
plt.title("Histogram of Student Marks")
# Show plot
plt.show()
```

HISTOGRAM OF IRIS SEPAL LENGTH

```
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
# Load the iris dataset
iris = load_iris()
# Sepal length is in column 0 of iris.data
sepal_length = iris.data[:, 0]
# Create histogram
plt.hist(sepal_length, bins=10, edgecolor="black")
# Labels and title
plt.xlabel("Sepal Length (cm)")
plt.ylabel("Frequency")
plt.title("Histogram of Sepal Length (Iris Dataset)")
# Show plot
plt.show()
```

SCATTER PLOT OF ROLL NUMBERS VS MARKS

```
import matplotlib.pyplot as plt
# Given data
rollnos = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]
marks = [22, 87, 5, 43, 56, 73, 55, 54, 11, 20, 51, 5, 79, 31, 27]
# Create scatter plot
plt.scatter(rollnos, marks, color="blue", marker="o")
# Labels and title
```

```
plt.xlabel("Roll Number")  
plt.ylabel("Marks")  
plt.title("Scatter Plot of Roll Numbers vs Marks")  
# Show plot  
plt.show()
```